Rank-controlled projective-incidence repair by weighted chart transport

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Abstract

Let two arbitrary functions map the points and hyperplanes of canonical $\mathrm{PG}(n, q)$ to those of $\mathrm{PG}(n, r)$, where $n \geq 2$ and q, r are prime powers. Let T count broken source flags, let d be the source incidence degree, and let ρ be the fraction of ordered projective frames that survive under the point map. If $\rho > 0$ and the flag error fraction δ is at most $\rho/(10^9 n^2)$, both functions agree, apart from at most $5T/q^{n-1} \leq 10T/d$ entries in total, with one coupled field-semilinear embedding. Neither input map is assumed injective. We give the weighted plane correction, its transport to higher rank, and the paired incidence gap used to obtain this estimate. The repair theorem and its two stated formal corollaries have existing Lean verification. This is a separate incidence-rigidity direction associated with the research on Erdős Problem 85; it does not settle that problem's eventual monotonicity question. Public priority remains unresolved.

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1 Statement and scope

Let P and $\mathcal{H}$ be the point and hyperplane sets of canonical $\mathrm{PG}(n,q)$. Throughout, source and target have the same projective dimension $n \geq 2$, while q and r are arbitrary prime powers. Put

$$N = \frac{q^{n+1}-1}{q-1}, \quad d = \frac{q^n-1}{q-1}, \quad \lambda = \frac{q^{n-1}-1}{q-1}, \quad a = d - \lambda = q^{n-1}, \quad m = Nd. \tag{1}$$

Consider arbitrary fixed functions

$$\phi : P \longrightarrow P(\mathrm{PG}(n,r)), \qquad \psi : \mathcal{H} \longrightarrow \mathcal{H}(\mathrm{PG}(n,r)).$$

A source flag is a pair (p,H) with $p \in H$. It is broken when $\phi(p) \notin \psi(H)$. Let T count these original flags once each and put $\delta = T/m$. An ordered projective frame is an ordered list of $n+2$ points, every $n+1$ of which are independent. Let ρ be the fraction of uniform ordered source frames whose point images form target frames. Repeated image points fail this condition.

For a field embedding $\sigma : \mathbb{F}_q \hookrightarrow \mathbb{F}_r$ and $A \in \mathrm{GL}_{n+1}(\mathbb{F}_r)$, write G for the point map $[v] \mapsto [A\sigma(v)]$ together with its coupled hyperplane map. The image of a hyperplane is the $\mathbb{F}_r$-linear span of the images of its source points. Thus it need not equal their set. Define

$$D(G) = |\{p : \phi(p) \neq G(p)\}| + |\{H : \psi(H) \neq G(H)\}|.$$

Theorem 1 (Coupled repair)**.** *If $\rho > 0$ and $\delta \leq \rho/(10^9 n^2)$, a field embedding $\mathbb{F}_q \hookrightarrow \mathbb{F}_r$ exists, and some coupled G satisfies*

$$D(G) \leq \frac{5T}{q^{n-1}} \leq \frac{10T}{d}. \tag{2}$$

Corollary 2 (Collapse or repair)**.** *Every pair (ϕ,ψ) satisfies at least one alternative:*

$$\textit{some coupled } G \textit{ has } D(G) \leq 10T/d, \qquad \textit{or} \qquad \rho \leq 10^9 n^2 \delta.$$

Corollary 3 (Incompatible fields)**.** *If there is no embedding $\mathbb{F}_q \hookrightarrow \mathbb{F}_r$, then $\rho \leq 10^9 n^2 \delta$ for every pair of maps.*

The classical finite-field criterion makes the last hypothesis explicit: the characteristics differ, or $q = p^u, r = p^v$ with $u \nmid v$. The first corollary follows because failure of its second alternative implies the hypotheses of Theorem 1. The second then follows because its repair alternative would supply a field embedding.

2 A weighted protected-plane lemma

First work in $\mathrm{PG}(2,q)$ and $\mathrm{PG}(2,r)$. Suppose the point and line maps fix, in the corresponding source and target coordinates, the seven points

$$U = [1:0:0], \quad V = [0:1:0], \quad O = [0:0:1], \quad I = [1:1:1],$$
$$W = [1:1:0], \quad E = [1:0:1], \quad J = [0:1:1]$$

and the six lines ℓ_∞, $d_0 : y = x$, $v_0 : x = 0$, $v_1 : x = 1$, $h_0 : y = 0$, $h_1 : y = 1$. Write (x,y) for $[x:y:1]$, and set

$$v_x : X = x, \quad h_y : Y = y, \quad d_b : Y = X + b, \quad \ell_a : Y = aX, \quad \ell_{a,b} : Y = aX + b.$$

Sums	Parameters	Flags
2	$x; y$	(V, v_x); (U, h_y)
2	x, y	$((x, y), v_x)$; $((x, y), h_y)$
2	t	$((t, t), v_t)$; $((t, t), h_t)$
1	t	$((t, t), d_0)$
1	$a \neq 0$	(O, ℓ_a)
1	$a \neq 0, t$	$((t, at), \ell_a)$
2	$a \neq 0$	$((1, a), v_1)$; $((1, a), h_a)$
2	$a \neq 0$	$((a^{-1}, 1), v_{a^{-1}})$; $((a^{-1}, 1), h_1)$
2	$a \neq 0$	$((1, a), \ell_a)$; $((a^{-1}, 1), \ell_a)$
1	b	(W, d_b)
1	x, b	$((x, x + b), d_b)$
4	$b; i = 0, 1$	$((i, i + b), v_i)$; $((i, i + b), h_{i+b})$, for each fixed i
2	$b; i = 0, 1$	$((i, i + b), d_b)$, for each fixed i
1	a, b, x	$((x, ax + b), \ell_{a,b})$

Table 1: The 24 weighted flag families.

Let $b(p, \ell)$ be the broken-flag indicator. The cost η is the sum of the 24 normalized sums listed in Table 1. A family with k varying field parameters has denominator q^k. Restricted nonzero parameters retain this denominator, rather than replacing q by $q - 1$. Each displayed flag in a row is a separate sum. Repetitions are intentional.

Define $B_X = \{x : (V, v_x) \text{ is broken}\}$, $X = \{x \notin B_X : \psi(v_x) \neq \ell_\infty\}$, and $X_0 = \mathbb{F}_q \setminus (B_X \cup X)$. Define B_Y, Y, Y_0 analogously with (U, h_y). For $x \in X$, the image vertical has equation $X = f(x)$; for $y \in Y$, the image horizontal has equation $Y = g(y)$. Extend f, g by zero on all other inputs. The protected values imply

$$0, 1 \in X \cap Y, \qquad f(0) = g(0) = 0, \qquad f(1) = g(1) = 1. \tag{3}$$

Lemma 4 (Weighted plane correction)**.** *If $\eta \leq 10^{-7}$, there is a field embedding $\sigma : \mathbb{F}_q \hookrightarrow \mathbb{F}_r$ with*

$$q - |X| \leq 17\eta q, \qquad q - |Y| \leq 17\eta q,$$
$$|\{x : f(x) \neq \sigma(x)\}| \leq 288\eta q, \qquad |\{y : g(y) \neq \sigma(y)\}| \leq 648\eta q.$$

The affine point map equals $(x, y) \mapsto (\sigma(x), \sigma(y))$ outside at most $971\eta q^2$ points. The conclusion also holds when the cost uses any nonnegative pointwise majorant of b.

2.1 Support estimates

Let Q be the affine points with a broken vertical or horizontal coordinate flag. In this subsection only, write $x = |X|$, $y = |Y|$, $b_X = |B_X|$, and $b_Y = |B_Y|$. Outside Q, intersection of the two image coordinate lines gives image $(f(u), g(v))$ for $(u, v) \in X \times Y$, image V on $X \times Y_0$, and image U on $X_0 \times Y$. Let Q_{diag} count diagonal points in Q, and let e_{diag} count broken affine point–d_0 flags. Since the target diagonal contains neither U nor V,

$$|X \triangle Y| \leq b_X + b_Y + Q_{\text{diag}} + e_{\text{diag}} \leq \eta q. \tag{4}$$

The two diagonal coordinate families dominate the union count in this inequality. For $a \neq 0$ put $r_a = |\{t \in X : at \in Y_0\}|$ and $s_a = |\{t \in X_0 : at \in Y\}|$. Let e_O count broken (O, ℓ_a) flags and E_{pen} broken $((t, at), \ell_a)$ flags. If the O flag is good, the image line contains O and cannot contain both U, V. Thus at least $\min(r_a, s_a)$ mixed points lie in Q or have a broken pencil-line flag. Each such point has two nonzero

coordinates and a unique nonzero slope. For a broken O flag use the upper bound q. Hence

$$\sum_{a\neq 0} \min(r_a, s_a) \le qe_O + |Q| + E_{\text{pen}} \le \eta q^2. \tag{5}$$

Moreover $|r_a - s_a| \le |x-y| + b_X + b_Y \le 2\eta q$. It follows that $\sum_{a\neq 0} r_a \le 3\eta q^2$. Counting the unique slope for each pair in $(X \setminus \{0\}) \times Y_0$ gives

$$(x-1)(q-y-b_Y) \le 3\eta q^2, \qquad (x-1)(q-y) \le 4\eta q^2. \tag{6}$$

For $a \in Y_0 \setminus \{0\}$ with $a^{-1} \in X_0$, use the two points $(1,a)$ and $(a^{-1},1)$ on ℓ_a. Their images are V, U unless a coordinate flag fails. If those and both point-line flags and the O flag are good, one target line contains O, U, V, a contradiction. Let R_Q count appearances in Q in these two families and R_F their broken ℓ_a flags. The table pays for each of these counts. All other nonzero a lie in $Y \cup B_Y$ or have inverse in $X \cup B_X$. Therefore

$$x + y \ge q + 1 - (b_X + b_Y + e_O + R_Q + R_F) \ge q + 1 - \eta q. \tag{7}$$

There is no reciprocal exception at $a = 1$, by (3). For $q = 2$ that equation already gives $X = Y = \mathbb{F}_q$. For $q \ge 3$, (4) and (7), with $\eta \le 1/12$, give $x - 1 \ge (q - 1 - 2\eta q)/2 \ge q/4$. Equation (6) then proves $q - y \le 16\eta q$ and $q - x \le 17\eta q$. These are inequalities on integer counts with no rounding remainder.

2.2 Additive correction and protected values

The product identity $\phi(u,v) = (f(u), g(v))$ fails at at most

$$K \le q(q - |X|) + q(q - |Y|) + |Q| \le 35\eta q^2$$

affine points. Define $c(b)$ as the target intercept of $\psi(d_b)$ if its W flag is good and its image is a proper affine line; otherwise set $c(b) = 0$. In the proper case its slope is 1. In particular $c(0) = 0$. Let R count failures of $g(u + b) = f(u) + c(b)$. Each product exception is used once, a broken W flag at most q times, and a broken point–d_b flag once. Thus, writing e_W, E_{diag} for the latter counts,

$$R \le K + qe_W + E_{\text{diag}} \le 36\eta q^2, \qquad r_0 = R/q^2 \le 36\eta. \tag{8}$$

For each fixed $i = 0, 1$, the protected vertical line gives the sharper row bound

$$|\{b : g(i + b) \neq f(i) + c(b)\}| \le 20\eta q. \tag{9}$$

Indeed only missing horizontal coordinates, coordinate-flag failures on that column, W flags and its listed point–d_b flags need be charged. The table and the support bounds give at most $17\eta q$; we retain 20. The diagonal families likewise give

$$|\{t : g(t) \neq f(t)\}| \le 36\eta q. \tag{10}$$

We supply the additive correction argument for the cross-field codomain. Suppose $A : \mathbb{F}_q \to \mathbb{F}_r$ fails additivity on at most a fraction $\theta < 1/6$ of pairs. Put $z_x(y) = A(x + y) - A(y)$. For fixed x, the additive equations at $(x + y, z)$ and $(y, x + z)$ imply $z_x(y) = z_x(z)$ except on at most a fraction 2θ of independent (y, z). Hence z_x has a unique mode $H(x)$ of mass at least $1 - 2\theta$. The identity

$$z_{x+t}(y) = z_x(t + y) + z_t(y)$$

and the three mode bounds prove additivity of H, since $6\theta < 1$. If $A(x) \neq H(x)$, at least $1 - 2\theta$ of the additive equations in that row fail. Summing rows shows that A differs from H on at most $2\theta q$ inputs.

Apply the Pexider equation to $(a+x, b+y)$, $(a+x+y, b)$, $(a, b+y)$, and $(a+y, b)$. Each pair is uniform for independent uniform a, b, x, y. Subtraction cancels the g, c terms; averaging chooses a for which $A(x) = f(a+x) - f(a)$ has additive defect at most $4r_0$. The preceding correction supplies an additive H and a constant α such that $f = H + \alpha$ outside at most $8r_0 q$ inputs. Substitution into the original equation gives

$$g(z) - H(z) = \alpha + c(b) - H(b)$$

outside a fraction $9r_0$ of independent pairs (z, b). The largest atom β of $c(b) - H(b)$ has mass at least $1 - 9r_0$. Consequently

$$\begin{aligned} f &= H + \alpha && \text{outside at most } 8r_0 q \text{ inputs,} \\ g &= H + \alpha + \beta && \text{outside at most } 18r_0 q \text{ inputs,} \\ c &= H + \beta && \text{outside at most } 9r_0 q \text{ inputs.} \end{aligned} \tag{11}$$

For the second estimate use $9r_0/(1-9r_0) \le 18r_0$. All required inequalities, including $4r_0 < 1/6$ and $9r_0 < 1/2$, follow from $\eta \le 10^{-7}$. For each $i = 0, 1$, (9) and (11) exclude at most $(20\eta + 27r_0)q \le 992\eta q < q$ choices of b. A remaining choice gives $f(i) = H(i) + \alpha$. Equation (3) therefore gives $\alpha = 0$ and $H(1) = 1$. Equation (10) and the f, g exceptions exclude at most $(36\eta + 26r_0)q \le 972\eta q < q$ choices of t, so $\beta = 0$. Both corrected coordinate functions are H.

2.3 All slopes force one field embedding

The affine point map equals $(H(x), H(y))$ outside at most

$$Z_2 \le K + 26r_0 q^2 \le 971\eta q^2$$

points. Mark a triple (a, b, x) if $(x, ax+b)$ is a point exception or its flag on $\ell_{a,b}$ is broken. Each exceptional affine point occurs q times. The final table family gives a total

$$M \le qZ_2 + \eta q^3 \le 972\eta q^3 < q^3/4. \tag{12}$$

Let

$$K_0 = \{a \in \mathbb{F}_q : \text{some } s \in \mathbb{F}_r \text{ satisfies } H(ax) = sH(x) \text{ for every } x\}.$$

The scalar is unique because $H(1) = 1$. If $a \notin K_0$, no target line contains more than $q/2$ entries of the list $(H(x), H(ax+b))$, $x \in \mathbb{F}_q$. A vertical line gives a fiber of the nonzero additive map H; a nonvertical line gives a fiber of the nonzero additive map $H(ax) - sH(x)$; the infinity line contains none. Every fiber of a nonzero additive map has size at most $q/2$, including when the displayed point list has repetitions. Every unmarked entry lies on the actual image of $\ell_{a,b}$. Thus each $a \notin K_0$ and each b require at least $q/2$ marks, whence

$$|\mathbb{F}_q \setminus K_0| \le 2M/q^2 < q/2.$$

The set K_0 is a subfield. It contains 0 and 1, and scalar identities give closure under addition and multiplication. The scalar of a nonzero $a \in K_0$ is nonzero, since multiplication by a is surjective and H is nonzero; substituting $a^{-1}x$ gives inverse closure. A proper subfield has at most $q/2$ elements as a proper additive subgroup. Hence $K_0 = \mathbb{F}_q$. Its unique scalars define an injective unital field homomorphism σ, and $H(x) = \sigma(x)H(1) = \sigma(x)$. Equation (11) gives the constants 288 and 648. This proves Lemma 4. All estimates charge nonnegative flag errors, so a pointwise majorant can replace the indicator.

3 Protection of a higher-rank frame

Normalize a source frame to $E_0 = [e_0], \dots, E_n = [e_n], I = [e_0 + \cdots + e_n]$. If it survives, normalize its target images independently to that same coordinate frame. The first $n+1$ representatives form a basis; all coefficients of the last representative are nonzero. Rescaling that basis makes those coefficients 1, also in characteristic 2. Set

$$H_0 : X_0 = 0, \qquad H_i(0) : X_i = 0, \qquad H_i(1) : X_i = X_0 \quad (1 \le i \le n).$$

Protect H_0 and the n hyperplanes $H_i(0)$ by their n basis-point flags each. Protect $H_i(1)$ by the flag at I and its $n-1$ basis-point flags outside $\{0, i\}$. These $2n^2 + n$ original flags force the named hyperplane images, because each contains n independent named target points. For $j = 2, \dots, n$, let A_j be the coordinate plane spanned by E_0, E_1, E_j, and let B_j be the vector span of the complementary e_k. Its vector dimension is $n-2$; for $n = 2$ it is zero. Add the following protections, with $C_j = \{0, \dots, n\} \setminus \{0, 1, j\}$:

1. For $S_j = [e_0 + e_1 + e_j]$, protect its flags with $H_k(0)$ for $k \in C_j$ and with $H_1(1), H_j(1)$. These n target hyperplanes intersect in the specified point.
2. For $D_j : X_j = X_1$, protect its flags at E_0, S_j and E_k for $k \in C_j$. Their n images are independent and force D_j.
3. For $W_j = [e_1 + e_j]$, protect its flags with H_0, D_j and the $H_k(0)$ for $k \in C_j$.
4. For $E_j^* = [e_0 + e_1]$, use $H_j(0), H_1(1)$ and those $H_k(0)$.
5. For $J_j^* = [e_0 + e_j]$, use $H_1(0), H_j(1)$ and those $H_k(0)$.

The last three groups again define unique intersections of n target hyperplanes. They cost $5n$ flags per plane. Repeated flags and points remain counted slots. The total is

$$M_{\text{cert}} = 2n^2 + n + 5n(n-1) = 7n^2 - 4n \le 7n^2. \tag{13}$$

Call a surviving frame protected if all these original flags are good. For each j, this forces the seven points and six extended lines of Section 2, with E_1, E_j, E_0 corresponding to U, V, O. No other point-image injectivity is imposed.

4 Transport and its unconditional load

Fix a protected frame. Let A'_j, B'_j be its corresponding target subspaces. A line ℓ in A_j extends to the source hyperplane $\widehat{\ell} = \text{span}(B_j, \ell)$. Define an induced point map by retaining $\phi(p)$ when it lies in A'_j, and replacing it by E_0 otherwise. Define the induced line image to be $\psi(\widehat{\ell}) \cap A'_j$ when $\psi(\widehat{\ell})$ contains B'_j, and the target infinity line otherwise. In the first case the intersection is a line: a hyperplane containing both B'_j and A'_j would contain their direct sum, the whole target vector space. Write b_{orig} for original broken-flag indicators, and put

$$\begin{aligned} U_j(p, \ell) = b_{\text{orig}}(p, \widehat{\ell}) &+ \sum_{k \in C_j} b_{\text{orig}}(p, H_k(0)) \\ &+ \sum_{k \in C_j} b_{\text{orig}}(E_k, \widehat{\ell}). \end{aligned} \tag{14}$$

These are $2n-3$ original flags, counted with multiplicity. If all are good, the middle group forces $\phi(p) \in A'_j$, the last group forces $B'_j \subseteq \psi(\widehat{\ell})$, and the first forces the induced incidence. Thus U_j pointwise majorizes induced broken flags. In particular it pays for both possible failures of restriction. Let η_j be the 24-family cost evaluated on U_j.

For $i = 1, \dots, n$ use the original global pencil $H_i(t) : X_i = tX_0$. Call t proper if $\psi(H_i(t))$ contains all E_k with $k \notin \{0, i\}$ and is not H_0. Its equation is then $X_i = f_i(t)X_0$; define $f_i(t) = 0$ otherwise. Protection gives proper inputs 0 and 1 with the usual values. In the induced plane A_j, the proper vertical set and its extended function are exactly this global set for $i = 1$ and f_1. The horizontal counterparts are exactly those for $i = j$ and f_j. Indeed, missing complementary containment sends the induced line to infinity. If complementary containment holds, the local infinity-anchor flag supplies precisely the remaining basis direction. The intersection is the local infinity line exactly when its span with B'_j is H_0. The extensions by zero therefore also agree at every nonproper input.

Now choose a uniform ordered source frame F, before imposing any condition on its image. Source projective linear transformations act transitively on frames and flags. Every certificate slot and every original flag in (14), at each fixed table parameter, is an equivariant construction from F. Each such slot is consequently uniform on the m source flags and has expected error δ. Equivalently each fixed slot takes any one source flag exactly F_{count}/m times. This identity precedes conditioning on survival. The raw source formulas for U_j and η_j make sense for every frame, even when its target image does not survive. Each normalized table sum has mass at most 1. Hence

$$\mathbb{E}\eta_j \le 24(2n-3)\delta.$$

For $p_x = [e_0 + x_1e_1 + \cdots + x_ne_n]$, define

$$\gamma(F) = q^{-n} \sum_{x \in \mathbb{F}_q^n} \sum_{i=1}^{n} b_{\text{orig}}(p_x, H_i(x_i)), \qquad \mathcal{E}(F) = \gamma(F) + \sum_{j=2}^{n} \eta_j(F).$$

The same equivariance gives $\mathbb{E}\gamma = n\delta$ and

$$\mathbb{E}\mathcal{E} \le [n + 24(n-1)(2n-3)]\delta \le 48n^2\delta. \tag{15}$$

The certificate fails with probability at most $7n^2\delta$. Markov's inequality and (15) show that the probability of certificate failure or $\mathcal{E} > 10^{-7}$ is at most

$$(7 + 48 \cdot 10^7)n^2\delta < 5 \cdot 10^8 n^2\delta \le \rho/2.$$

Since survival has probability $\rho > 0$, some surviving protected frame has $\mathcal{E} \le 10^{-7}$. If $\delta = 0$, every raw nonnegative cost is zero and the same conclusion follows directly. No conditional uniformity or division by zero is needed.

5 Coherence and reconstruction of the affine chart

Fix that frame. Lemma 4 supplies, for each j, an embedding $\sigma_j : \mathbb{F}_q \hookrightarrow \mathbb{F}_r$ with

$$|\{t : f_1(t) \ne \sigma_j(t)\}| \le 288\eta_j q, \qquad |\{t : f_j(t) \ne \sigma_j(t)\}| \le 648\eta_j q.$$

There are at most $17\eta_j q$ nonproper inputs in either pencil. For two charts the embeddings disagree on at most $288(\eta_j + \eta_k)q \le 288 \cdot 10^{-7} q < q/2$ inputs, since both approximate the same original function f_1. Distinct additive maps can agree on at most $q/2$ inputs, by the kernel bound for their difference. Thus all σ_j equal one σ. For $n = 2$ there is just one chart. Write $\eta_\Sigma = \sum_{j=2}^{n} \eta_j$. Use chart 2 once for coordinate 1 and chart j for

coordinate j. Summing bounds, the number of nonproper coordinate inputs is at most $17(\eta_2+\eta_\Sigma)q \le 34\eta_\Sigma q$, and the number of scalar disagreements is at most $(288\eta_2 + 648\eta_\Sigma)q \le 936\eta_\Sigma q$. If all x_i are proper and all n coordinate flags at p_x are good, $\phi(p_x)$ is the unique intersection of $X_i = f_i(x_i)X_0$, $1 \le i \le n$. The union bound across the whole affine chart gives at most $(\gamma + 34\eta_\Sigma)q^n$ failures of this product identity. After correcting the scalar values, the point map agrees with $(x_i) \mapsto (\sigma(x_i))$ outside at most

$$Z \le (\gamma + 970\eta_\Sigma)q^n \le 971\mathcal{E}q^n. \tag{16}$$

This is agreement on the entire affine chart, rather than only on the coordinate planes. Extend this embedding projectively to G. Its image of H_0 already equals $\psi(H_0)$.

6 Both maps are close, including at infinity

Two intersection bounds apply also to proper subfield embeddings. A target hyperplane different from $G(H)$ contains at most λ images of source points of H. A target point different from $G(p)$ lies on at most λ images of source hyperplanes through p. To see the first, normalize G to coordinatewise σ. The restriction of the wrong target equation to the n-dimensional $\mathbb{F}_q$ vector space defining H is a nonzero σ-linear functional into $\mathbb{F}_r$. Its kernel has $\mathbb{F}_q$-dimension at most $n-1$. For the second, the $\mathbb{F}_q$ coefficient space of hyperplanes through p has dimension n and spans the corresponding target coefficient space over $\mathbb{F}_r$. Evaluation at a target point other than $\sigma(p)$ is nonzero on that space and has the same kernel bound. For an affine source hyperplane, a wrong target hyperplane contains at most q^{n-2} correctly embedded affine points. On its affine parameter space, the equation is either nonconstant, with fibers of size at most q^{n-2}, or a nonzero constant, with no zeros. A zero constant would force containment of the whole target-field span $G(H)$.

Let B count wrong affine source hyperplanes, and I_{bad} wrong points on H_0. A wrong affine hyperplane has $a = q^{n-1}$ affine points; at most a/q correctly mapped ones can preserve their flags. Charge an affine point exception on its at most d source hyperplanes:

$$(a - a/q)B \le T + dZ. \tag{17}$$

A wrong infinity point lies on $d-1$ affine hyperplanes, and at most λ correctly mapped ones can contain its image. Each wrong affine hyperplane contains λ infinity points. Thus

$$(a-1)I_{\text{bad}} \le T + \lambda B. \tag{18}$$

These are separate estimates and need not charge disjoint errors. Using $d/a \le 2$, $\lambda/(a-1) = 1/(q-1)$ and $a/(a-1) \le 2$ gives

$$D(G) = Z + B + I_{\text{bad}} \le 9Z + 6T/a. \tag{19}$$

Since $q^n \le N$, $T/a \le 2\delta N$, $\mathcal{E} \le 10^{-7}$, and $\delta \le 1/(10^9 n^2) \le 1/(4 \cdot 10^9)$, this yields

$$D(G)/N \le 9 \cdot 971 \cdot 10^{-7} + 12\delta < 1/1000 < 1/20. \tag{20}$$

Both original functions are now close to the same G.

7 The paired source-design gap

Let $S, \mathcal{B}$ be the wrong point and hyperplane sets, of sizes s, t, and put $D = s + t$. Let J count source incidences between these sets. A wrong point has at most λ correctly mapped incident hyperplanes that can contain its wrong image. There are therefore at least $as - J$ broken flags with wrong point and correct

hyperplane. The dual bound gives at least $at - J$ with correct point and wrong hyperplane. The two types are disjoint, so

$$T \geq aD - 2J. \tag{21}$$

Let M now denote the source incidence matrix. It has degree d; two distinct points share λ hyperplanes. Hence $MM^{\mathsf{T}} = a\,\mathrm{Id} + \lambda \mathbf{1}\mathbf{1}^{\mathsf{T}}$. For $n_H = |S \cap H|$ this gives

$$\sum_H (n_H - ds/N)^2 = as(1 - s/N).$$

Center the indicator of $\mathcal{B}$ and use Cauchy–Schwarz to obtain

$$J \leq dst/N + \sqrt{as(1-s/N)t(1-t/N)} \leq dD^2/(4N) + \sqrt{a}\,D/2.$$

Substitution in (21) gives

$$T \geq \left(a - \sqrt{a} - \frac{dD}{2N}\right) D.$$

Since $a \geq 2$, $\sqrt{a} \leq 3a/4$; also $d \leq 2a$. For $D \leq N/20$ the coefficient is at least $a - 3a/4 - 2a/40 = a/5$. Equation (20) therefore proves $D(G) \leq 5T/a \leq 10T/d$. Undoing the independent source and target normalizations gives exactly one map $[v] \mapsto [A\sigma(v)]$ and its coupled hyperplane action. Counts and entry distances are preserved. This completes the proof of Theorem 1.

8 Examples and limits

For $n = 2$, the complementary sums in (14) are empty and there is one chart. The support proof explicitly includes $q = 2$. Every exact coupled semilinear embedding, including a proper subfield embedding, has $T = 0$, $\rho = 1$, and $D = 0$; invertibility over $\mathbb{F}_r$ ensures survival of source frames. Constant incident point and hyperplane maps instead have $T = 0$ and $\rho = 0$. More generally, point values in a proper target subspace W and hyperplane values containing W can preserve every incidence while losing every full frame. The positive survival condition detects this rank collapse.

For a scale check, start with an exact embedding and move one source point p to the image of a different point b, retaining all other entries. Exactly λ of the d hyperplanes through p also contain b, so exactly $a = q^{n-1}$ flags break. The distance to the original embedding is 1, and frames avoiding p survive. This tests the normalization T/d, but proves neither sharp constants nor that the small-error hypothesis holds for every such parameter choice. The constants are not optimized and can require very large spaces before the hypothesis admits nonzero integer errors. The theorem asserts existence; it supplies no implemented decoder or complexity guarantee. It concerns canonical projective spaces, not arbitrary non-Desarguesian planes or a general building theorem.

9 Relation to earlier work and Erdős Problem 85

The original question concerns eventual monotonicity of the four-cycle minimum-degree threshold function. That question remains open here. This manuscript supplies no determination of $f(49)$ or new extremal bound for it. The problem statement and its discussion were read on 17 October 2026 from Bloom's problem page [1]. The page attributes the question to Erdős under its keys [Er93, p. 345], [Er94b], [Er95], and [Er96]. Those keys do not represent fresh reading of the original articles; the bibliography endpoints were unavailable in the saved source review.

The saved comparison read Theorem 2.3.1 and Remark 2.3.2 of Kollár–Lieblich–Olsson–Sawin [2], printed pages 36–38, from the author-hosted version whose last-modified header was 18 September 2021.

That theorem assumes injection, a common finite field, and $n > 3$, and measures failures of whole lines with specified excluded subspaces. The present theorem uses supplied hyperplane data and original point–hyperplane flags. No reduction establishing blanket dominance in either direction is given. The saved comparison also read the introduction, Theorem 1, Lemma 1 and Theorem 2 context, pages 1–4, of Pankov–Vetterlein [3], from the author-hosted version whose header was 23 December 2021. Its exact one-way orthogonality statements over finite-dimensional complex Hilbert spaces are not noisy finite-field two-map bounds. The complete proofs of these external works were not re-audited.

Kollár et al. explicitly credit Blum–Luby–Rubinfeld [4] for linearity testing. The additive correction above belongs to that classical tradition and is included directly to handle the required codomain and protected values. No claim is made to originate additive correction, semilinear geometry, or projective-design mixing. The source review also inspected only the abstract of Cardinali–Giuzzi–Pasini [5]; it establishes no subsumption comparison. Coverage of later citing work is incomplete. Further comparison may change the priority assessment.

10 Formal statements and review status

The accepted package records official theorem checks, full local import closure replay, zero holes, and only the axioms `propext`, `Classical.choice`, and `Quot.sound`. The exact principal mapping is as follows, in namespace `Erdos85.IncidenceRepair`:

- Theorem 1: `projective_incidence_repair`, in `Openmath/IncidenceRepair/Main.lean`, asserting `RepairClaim K L n`.
- Corollary 2: `repair_dichotomy`, in `Openmath/IncidenceRepair/Corollaries.lean`.
- Corollary 3: `frameSurvival_bound_of_no_field_embedding`, in the same file.

The numerical finite-field criterion is the classical interpretation of the formal no-embedding hypothesis. This edition reuses existing accepted evidence for unchanged mathematical sources. Compiling its typesetting does not check the mathematics, and this exposition has not received a new independent statement-by-statement review.

11 AI use

AI systems developed the mathematical argument, formalization and earlier source reviews, and prepared this exposition. Robert Huynh directed the research and is the sole named author. No human mathematical review is claimed. Formal verification concerns precise formal statements; it does not establish novelty or public priority. This working draft reorganizes existing work and makes no new mathematical claim.

References

[1] T. Bloom, *Erdős Problem 85*, https://www.erdosproblems.com/85, and discussion https://www.erdosproblems.com/forum/thread/85. Accessed 17 October 2026; statement page last edited 6 December 2025.

[2] J. Kollár, M. Lieblich, M. Olsson and W. Sawin, *The Zariski topology, linear systems, and algebraic varieties*, Theorem 2.3.1 and Remark 2.3.2, https://williamsawin.com/ReconstructionBook.pdf.

[3] M. Pankov and T. Vetterlein, *A geometric approach to Wigner-type theorems*, https://www.flll.jku.at/sites/default/files/u24/Wigner-type-theorems.pdf.

[4] M. Blum, M. Luby and R. Rubinfeld, Self-testing/correcting with applications to numerical problems, *Journal of Computer and System Sciences* 47(3) (1993), 549–595, https://doi.org/10.1016/0022-0000(93)90044-W.

[5] I. Cardinali, L. Giuzzi and A. Pasini, https://arxiv.org/abs/2306.17079. Abstract-only comparison in the saved source review.